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Barrowman Equations

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Introduction

Assessing the static pitch stability of a rocket is an important and contentious object. It has been seriously proposed that a cardboard profile of the rocket be made and its C.G. used as an estimate on the Center of Pressure. Another proposal would attach a string to the rocket at its C.G. When whirled around one's head we have a very poor man's wind tunnel. And, there are many more such cockamamie schemes, all found in the same dark and smelly place as these two.

For many years the amateur rocket community has considered the Barrowman Equations to the gold standard approach to estimating the center of pressure location. These use slender body loading for the body of revolution and Diederich's approximate lifting surface theory. As such, this model only applies to a subsonic single stage rocket. The full Barrowman thesis[1] can be found on Apogee Component's web site.

Discussion

The subsonic code documented here uses Munk's slender body theory[4] for the slender parts of the body of revolution. If the first entry in the body profile table of radius vs. station is not zero (pointed nose tip) at tangent hemispherical cap is fitted and a separate model for its normal force[5] is used. The stand alone fins are modeled using Diederich's approximate lifting surface model[2] including the full Prandtl-Glauert compressibility corrections. Full slender body estimates for the wing-body and body-wing interference[3] are used. For a two stage (two sets of fins) configuration, the vortex interference of the upper stage fins on the lower stage fins is modeled using vortex strengths[3] and strip theory first stage normal force to derive a correction factor[7].

There are two such files archived on this site: BarrowmanSub.xls (for subsonic) implements the original Barrowman equations except for a hemispherical nose cap and the full NACA Report 1307 interference corrections plus fin-fin vortex interference documented[7]. BarrowmanM.xls (M for high Mach No.) models a supersonic single stage rocket using second order shock expansion theory[6] for the body loading and lifting surface theory for the fins. A hemispherical nose tip[5] is also modeled using Lee's modified Newtonian aerodynamics.

References

1. Barrowman, J.S. and Barrowman, J.A., "The Theoretical Prediction of the Center of Pressure", Master's Thesis, Catholic University. A copy can be found online by Googling Barrowman equations, going to Model Rocket Stability, and then opening Barrowman's Original NARAM R&D Report

2. Diederich, F.W., “A Plan-Form Parameter for Correlating Certain Aerodynamic Characteristics of Swept Wings”, N.A.C.A. Technical Note 2335, April 1951.
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4. Munk, M.M., “The Aerodynamic Forces on Airship Hulls”, N.A.C.A. Report 184, 1924
5. Hoult, C.P., “Normal Force Acting on a Hemispherical Nose Tip”, RST memo, March 2010
6. Syvertson, C.A. and Dennis, D.H., “A Second-Order Shock-Expansion Method Applicable to Bodies of Revolution Near Zero Lift”, N.A.C.A. Report 1328, October 1955
7. Hoult, C.P., “N.A.C.A. Report 1307 Supplement for Downwash Dependent Fin-Fin Interference”, RST memo, April 2010